Sargassum Data Report

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Standalone summary of sargassum (resource_id=3) data currently in the ca-biositing database, across proximate, compositional, ICP, pretreatment, and fermentation analyses. XRF is omitted — no sargassum XRF records exist in the database.

Samples are color-coded consistently within each figure. Only QC-passed records are shown.

Underlying data for all figures below: `sargassum_report_data.csv`

Proximate Analysis

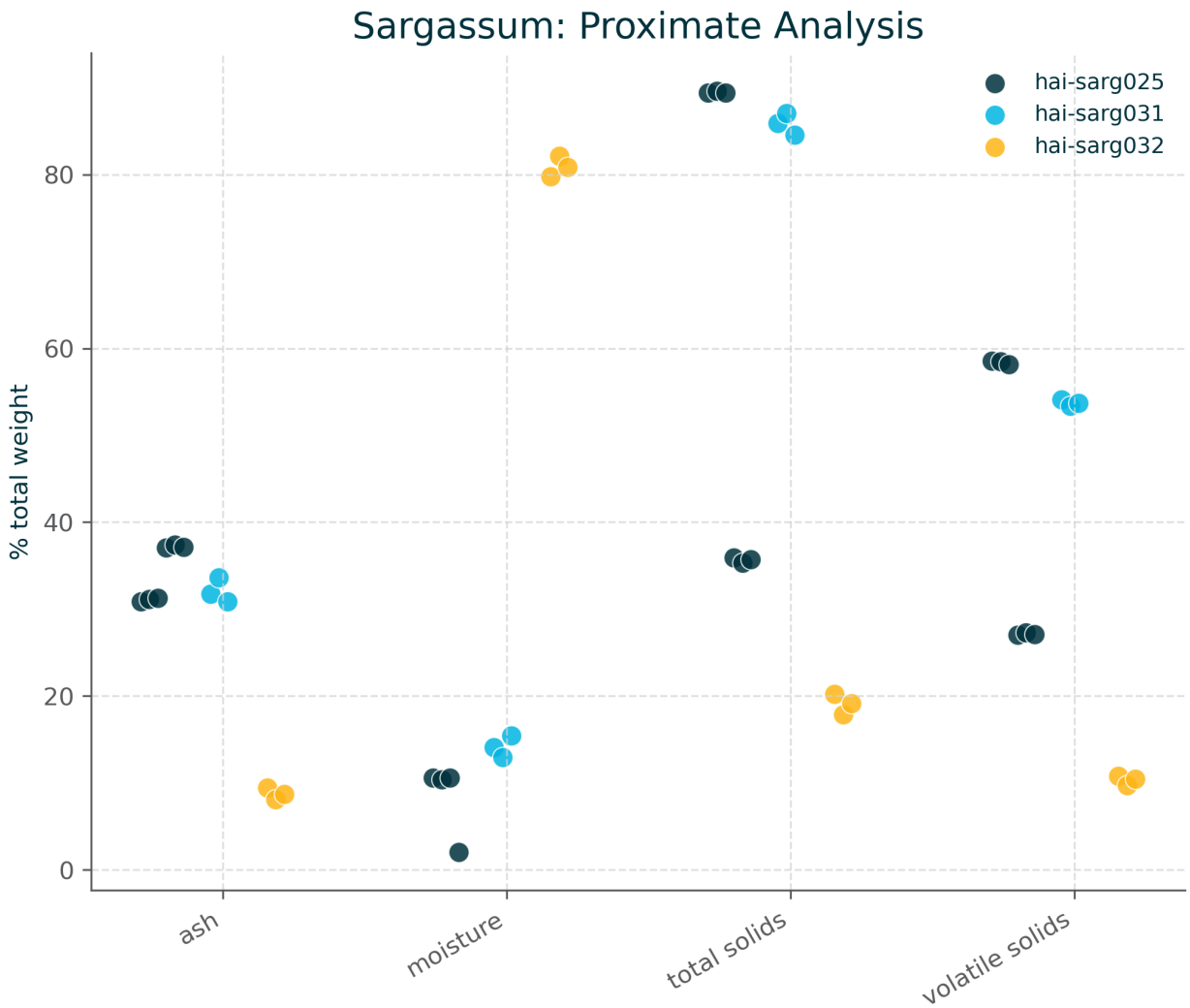


Figure 1: Proximate Analysis

Moisture, ash, total solids, and volatile solids content.

Sample	Records
hai-sarg025	22

Sample	Records
hai-sarg031	12
hai-sarg032	12

Compositional Analysis

Xylose, glucose, and acid insoluble residue content (other compositional parameters excluded from this report). Acid insoluble residue is the Klason-type residual fraction; for terrestrial plants this is conventionally reported as lignin, but sargassum (a brown alga) lacks true lignin, so it is labeled more precisely here.

Sample	Records
hai-sarg025	9

ICP Elemental Analysis

Elemental concentrations (ppm) measured by ICP-OES.

Sample	Records
hai-sarg025	26

Pretreatment Yields

Glucose and xylose yields from the only pretreatment method tested on sargassum (cholinium lysinate, 140°C).

Sample	Records
hai-sarg025	28

Fermentation Metrics

Sugar consumption, product titers, and yield metrics from fermentation runs.

Sample	Records
hai-sarg025	54
hai-sarg032	27

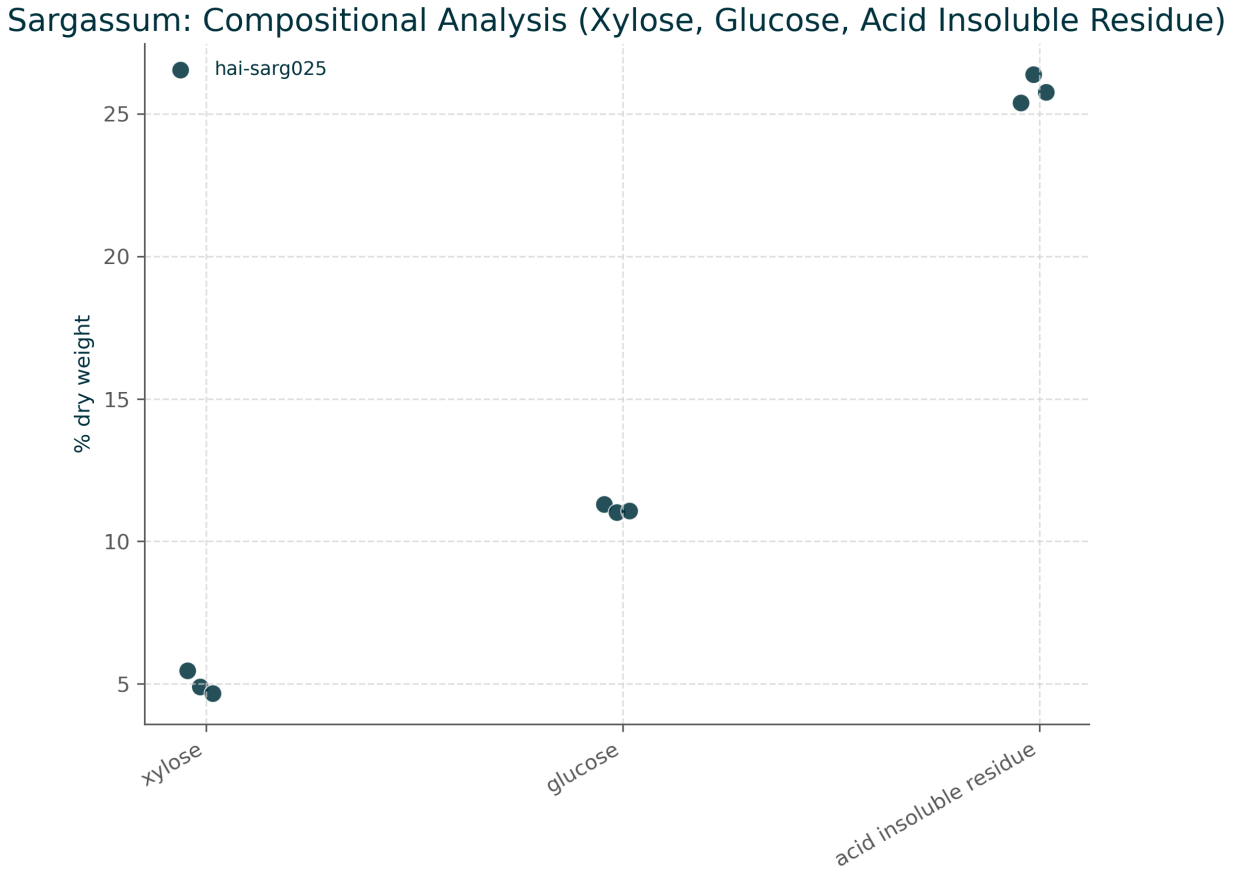


Figure 2: Compositional Analysis

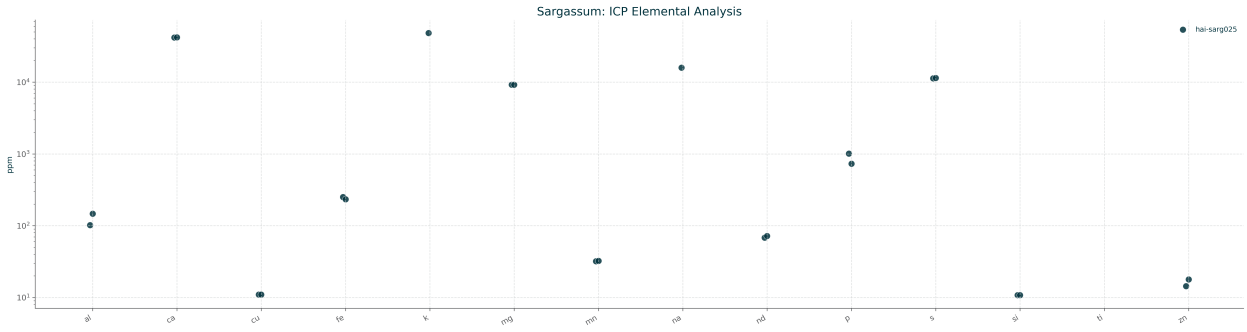


Figure 3: ICP Elemental Analysis

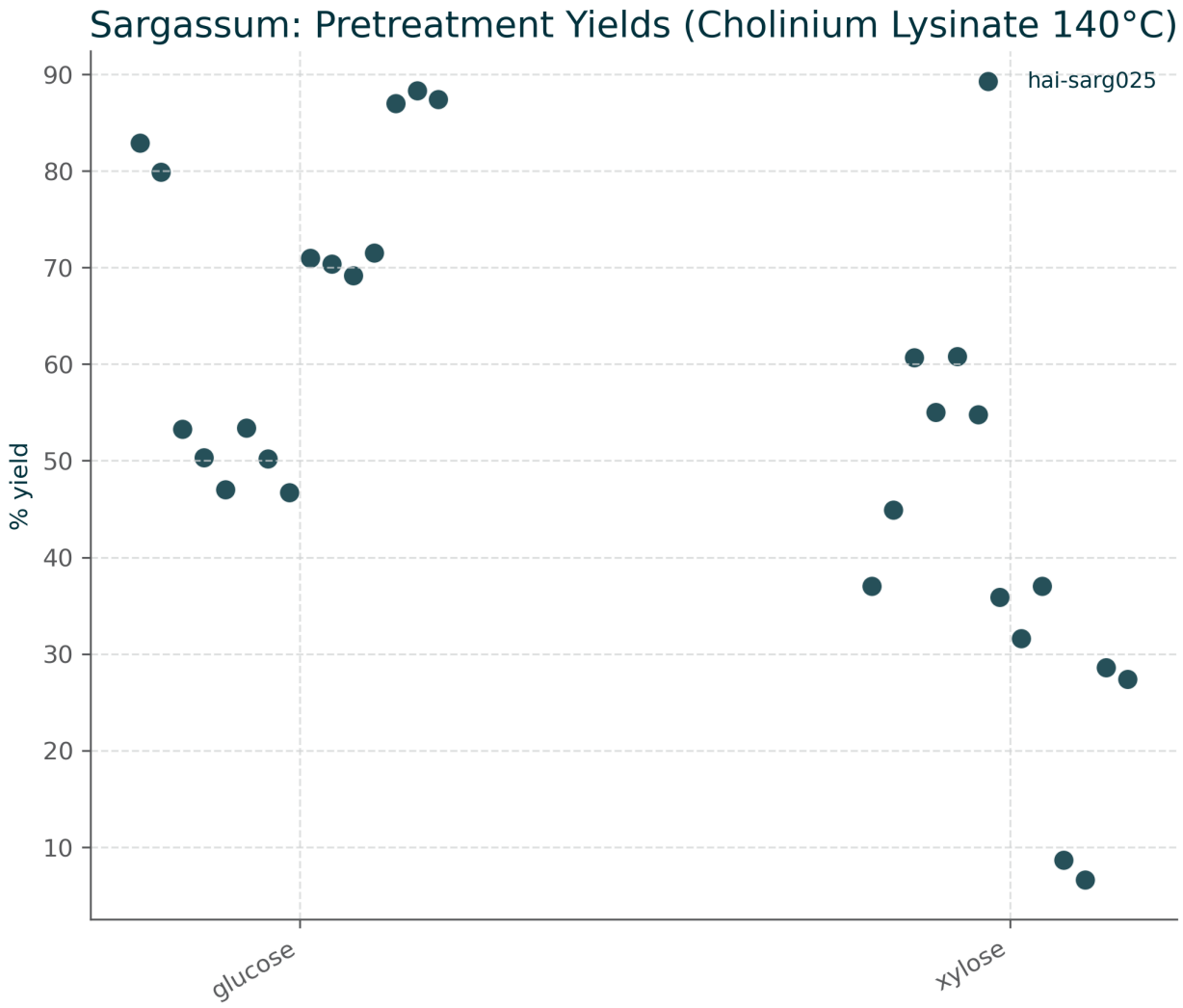


Figure 4: Pretreatment Yields

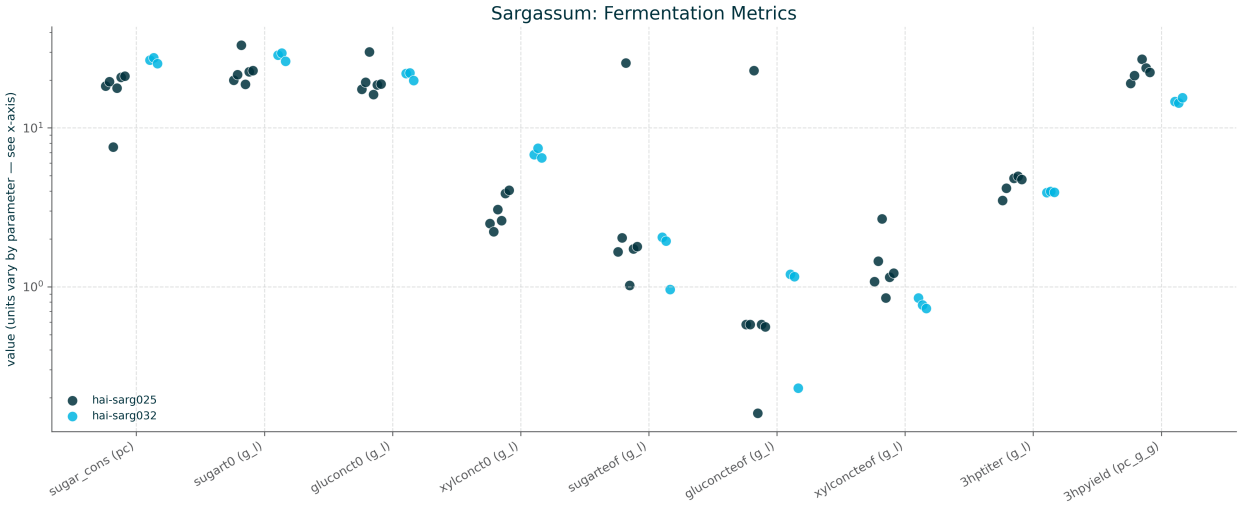


Figure 5: Fermentation Metrics